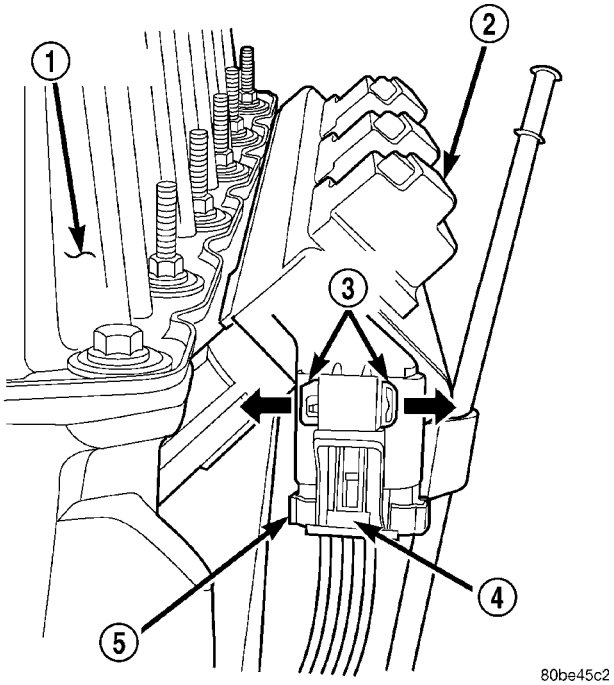

Fig. 27 Ignition Coil Electrical Connector—4.0L 6–Cylinder Engine

1 - REAR OF VALVE COVER
2 - COIL RAIL
3 - SLIDE TAB
4 - RELEASE LOCK
5 - COIL CONNECTOR

INSTALLATION

(1) Connect engine harness connector to coil by snapping into position. Move slide tab towards engine (Fig. 27) for a positive lock.

(2) Position ignition coil rubber boots to all spark plugs. Push down on coil assembly until bolt bases have contacted cylinder head

(3) Install 4 coil mounting bolts. Loosely tighten 4 bolts just enough to allow bolt bases to contact cylinder head. Do a final tightening of each bolt in steps down to 29 N·m (250 in. lbs.) torque. Do not apply full torque to any bolt first.

(4) Connect negative battery cable to battery.

IGNITION COIL - 2.4L

DESCRIPTION

The coil assembly consists of 2 different coils molded together. The assembly is mounted to the top of the engine (Fig. 28).

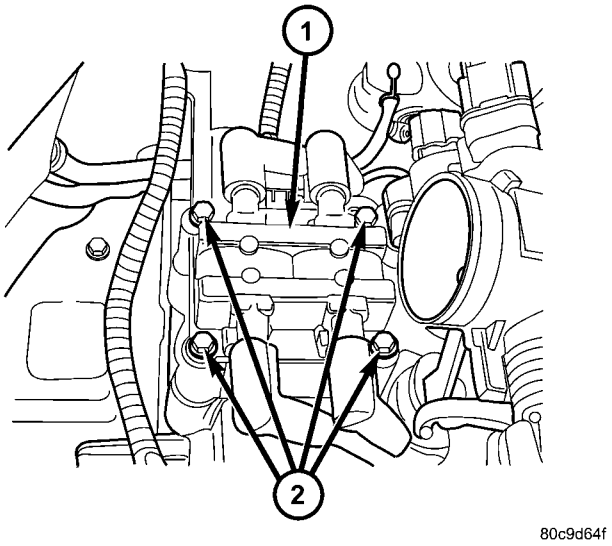

Fig. 28 IGNITION COIL - 2.4L

1 - IGNITION COIL
2 - MOUNTING BOLTS (4)

OPERATION

The coil fires two spark plugs simultaneously. One plug is under compression, the other plug fires on the exhaust stroke (lost spark). Coil number one fires cylinders 1 and 4, and coil number two fires cylinders 2 and 3.

The Auto Shutdown (ASD) relay provides battery voltage to the ignition coil. The PCM provides a ground contact (circuit) for energizing the coil(s). The PCM will de-energize the ASD relay if it does not receive the crankshaft position sensor and camshaft position sensor inputs.

Base ignition timing is not adjustable. By controlling the coil ground circuit, the PCM is able to set the base timing and adjust the ignition timing advance. This is done to meet changing engine operating conditions.

The ignition coil is not oil filled. The windings are embedded in an epoxy compound. This provides heat and vibration resistance that allows the ignition coil to be mounted on the engine.

Spark plug cables (secondary wires or cables) are used with the 2.4L engine.

REMOVAL

(1) Disconnect electrical connector at rear of coil.
(2) Remove all secondary cables from coil.
(3) Remove 4 coil mounting bolts (Fig. 29).
(4) Remove coil from vehicle.